

Memory Protection Integration

Defense-in-Depth Architecture

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- Independent guarantees from each layer
- Failure of any layer doesn't compromise security

Combined Security

- Cryptographic + Economic + Language + Runtime
- Multi-layered attack resistance

User Process

Request: Memory Access
UserCapability cap

Present capability

Blind Ledger

Cryptographic Access Control

ledger_verify(cap)

- Validate capability hash
- Check permissions
 - Verify epoch
 - Return entry

BLAKE2b-256 validation

Check budget

Pebble Accounting

Token Economics

pebble_check_budget()

- Check token availability
- Calculate PoW difficulty
- Verify work completed
 - Consume tokens

Economic throttling

Validate ownership

Borrow Checker

Ownership Tracking

borrow_check_ownership()

- Validate ownership
- Check borrowing rules
 - Verify no conflicts
 - Grant access

Rust semantics

Enforce access

Runtime Enforcement

Active Validation

borrow_checked_memmove()

- Validate actual access
- Check ownership again
- Monitor operations
- Panic on violation

Active protection

Perform access

Physical Memory

Actual 4KB Pages

Hardware memory access
Protected by all layers above

Fast Path Performance

- Hash table lookup: $O(1)$
- Bloom filter: $O(1)$
- Token validation: $O(1)$
- Runtime check: ~10ns overhead